Agri MLLM

A Multimodal Agricultural Assistant for Text, Image, and Audio Queries

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Talk Roadmap

- Problem and product vision
- Platform experience and supported modalities
- Model architecture and training pipeline
- Datasets, results, and example output
- Limitations and next steps

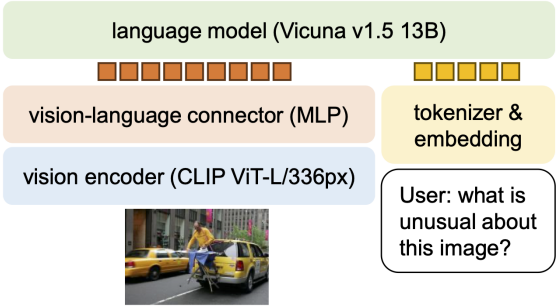
Problem We Are Solving

Core Need

Farmers, students, and agronomy practitioners need quick crop-health guidance without always typing long technical descriptions.

- Disease recognition often depends on both **visual cues** and **domain knowledge**.
- Many users prefer different input modes depending on context:
  - field image + typed question,
  - text-only agronomy question,
  - spoken question when hands are busy.
- The goal is one assistant that can unify these entry points.

Inspiration and Our Adaptation



Inspired by LLaVA-1.5

- CLIP vision encoder
- simple MLP vision-language connector
- efficient visual instruction tuning recipe

Adapted to our setting

- domain goal: agriculture, not general vision chat
- domain data: KisanVaani + PlantVillage
- compute constraints: Qwen2.5-0.5B, LoRA, Colab T4
- design choice: keep all 256 spatial patches for leaf-level detail

Inspired by *Improved Baselines with Visual Instruction Tuning*; figure extracted from `papers/2310.03744v2.pdf`.

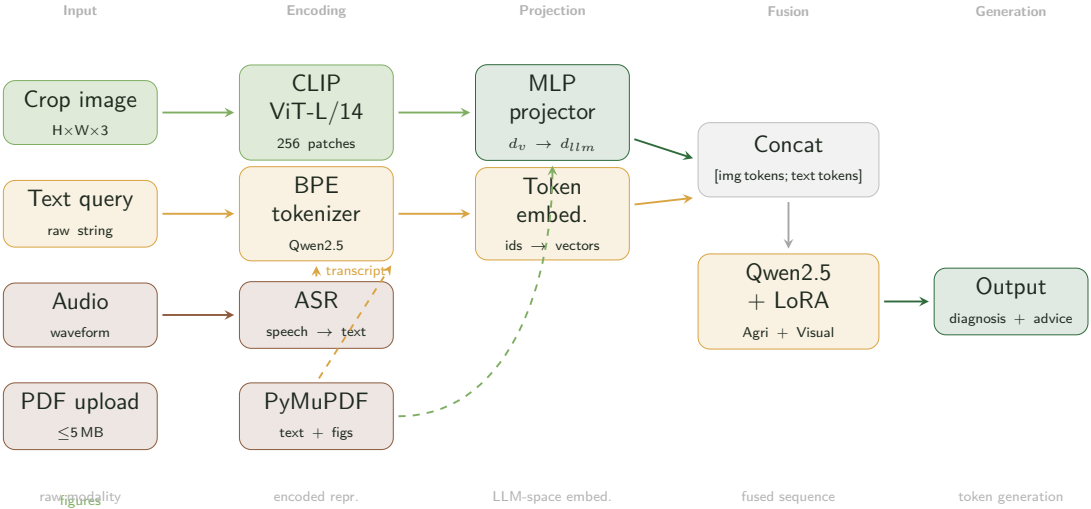
Platform Vision

A Multimodal ChatBot

- **Image + text**: "What disease is visible and how severe is it?"
- **Text only**: "How much phosphorus per hectare should I apply to wheat?"
- **Audio only**: spoken agronomy question captured by the platform
- **PDF files**

End-to-End System Pipeline

The Actual Model: From Input to Output



Training Pipeline in Three Steps

1. **Stage 0 Text-only domain adaptation**
   - Fine-tune a LoRA on agricultural QA text.
2. **Stage 1 visual alignment**
   - Freeze ViT and LLM.
   - Train only the projector on image-caption pairs.
3. **Stage 2 multimodal adaptation**
   - Keep ViT frozen.
   - Train projector + new visual LoRA on VQA triplets.

Why this works

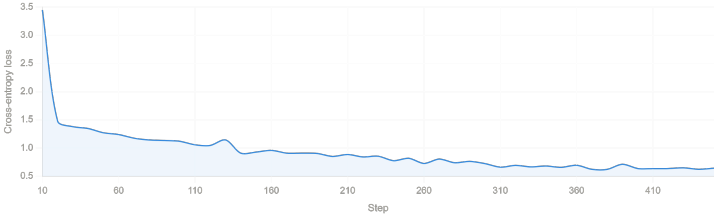
The pipeline first teaches the model agricultural language, then teaches image tokens to "speak" the LLM embedding space, then tunes multimodal reasoning.

Stage 0: Prepare the Agriculture-Aware LLM

Component	Details
Train	LoRA on base LLM
Data	Agriculture text corpus: KisanVaani/agriculture-qa-english-only
Loss	Causal LM (next token prediction)
Result	LLM that knows plant diseases, agronomic terminology

- Source in code: `KisanVaani/agriculture-qa-english-only`
- Processed rows in logs: **22,574 usable**; training JSONL subset: **8,000**
- Stage 0 setup: **450 max steps**, effective batch size **16**

Stage 0 Plot: LLM Fine-Tuning



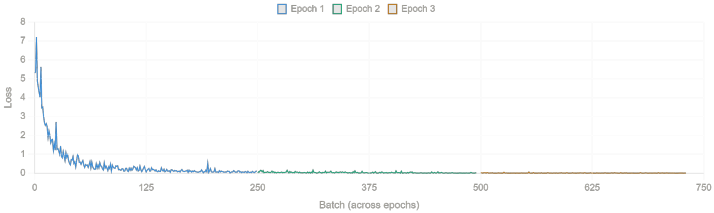
What changed

- LoRA trained on **8,000** agricultural QA samples
- Sharp early drop, then steady convergence
- Final logged train loss: **0.9349**

Stage 1: Train the Visual Projector

Component	Details
Freeze	ViT + LLM
Train	MLP projector only
Data	(image, caption) pairs from disease dataset (PlantVillage)
Loss	Causal LM on captions given image tokens
Result	Projector that speaks the LLM's embedding language
Purpose	Alignment stage — teach image tokens to look like text tokens

Stage 1 Plot: Caption Alignment



Interpretation

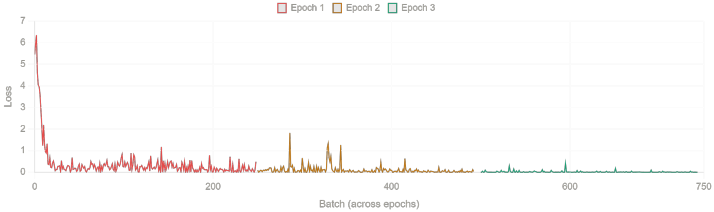
This is the cleanest convergence phase in the pipeline and shows that the projector successfully learns the vision-to-language bridge.

Stage 2: Fine-tune End-to-End with LoRA

Component	Details
Freeze	ViT
Train	MLP projector + new LoRA layers in LLM
Data	(image, question, answer) triplets — disease VQA (PlantVillage)
Loss	Causal LM on answers
Result	Full multimodal agriculture assistant

- Stage 2 set in logs: **1,000** generated VQA triplets from PlantVillage
- **4** QA pairs sampled per image from 5 templates

Stage 2 Plot: VQA Alignment



Interpretation

The first epoch is noisier because VQA is harder than captioning, but the run still converges cleanly by epoch 3.

Training Snapshot

Phase	Epoch / Run	Metric	Value
Text LoRA	final run	train loss	0.9349
Projector Stage 1	epoch 1	avg loss	0.5500
Projector Stage 1	epoch 2	avg loss	0.0347
Projector Stage 1	epoch 3	avg loss	0.0090
Stage 2 multimodal	epoch 1	avg loss	0.3996
Stage 2 multimodal	epoch 2	avg loss	0.0994
Stage 2 multimodal	epoch 3	avg loss	0.0141

- Report summary: **8.8M trainable parameters**, about **1.75%** of a 502M-parameter model.
- Stage 0 runtime reported: **1007 seconds** with about **7.15 samples/s**.

Audio Modality Integration

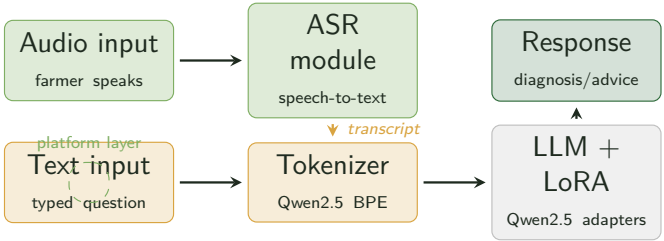
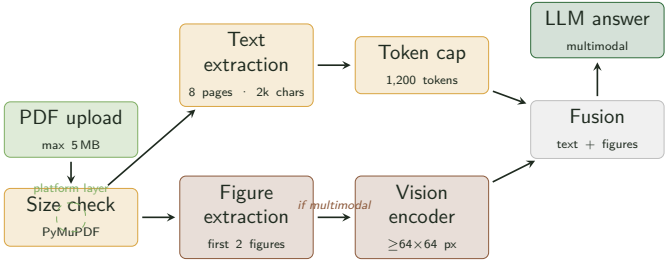


Figure: Audio follows the same inference path as typed text once transcribed.

Key points

- **ASR** lives in the platform layer not inside the model
- Transcript enters the **same tokenizer** as typed text
- Agriculture LoRA handles both typed and spoken queries unchanged
- Audio is a new **entry interface**, not a new reasoning pathway

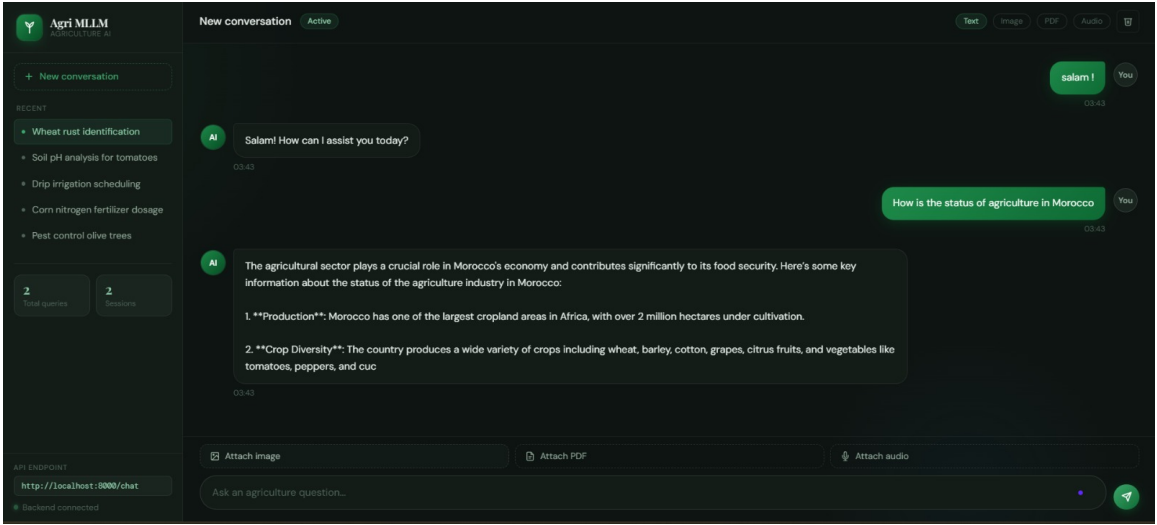
PDF Modality Integration



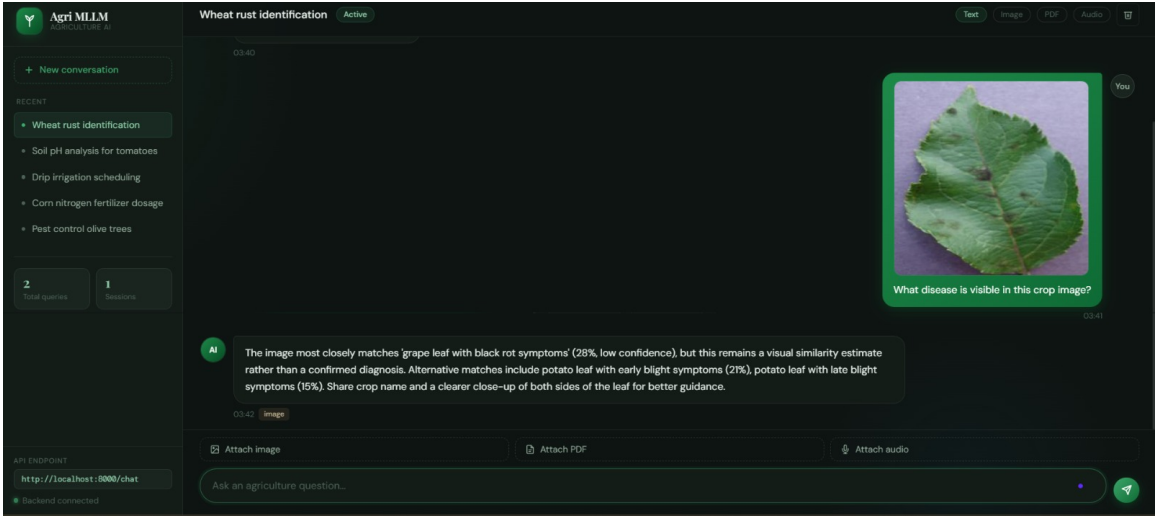
Key points

- PDFs capped at **5 MB**; parsed with PyMuPDF
- Text: first **8 pages**, 2,000 chars/page, **1,200 token** cap
- Figures: first **2 figures**, images $<64\times64$ px ignored
- Figures encoded by the **vision encoder** and passed through the PDF figure projector
- Text and figure tokens **fused** with the user question before the LLM

Demo - Text Only

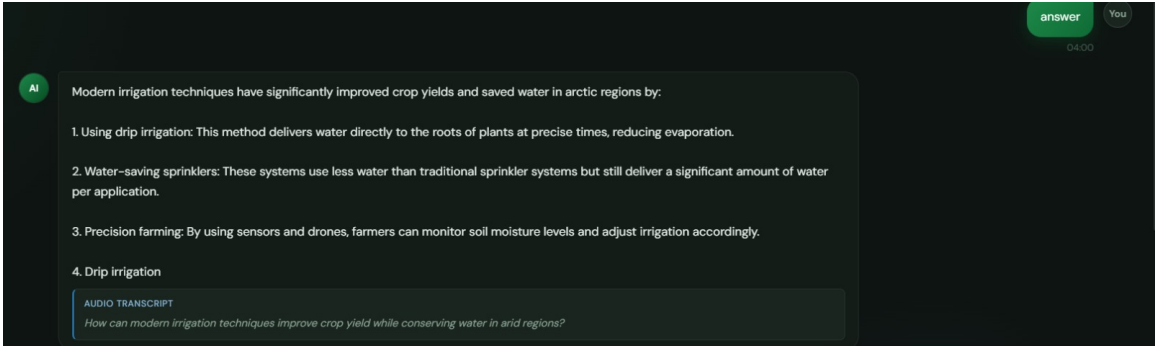


Demo - Image & Text

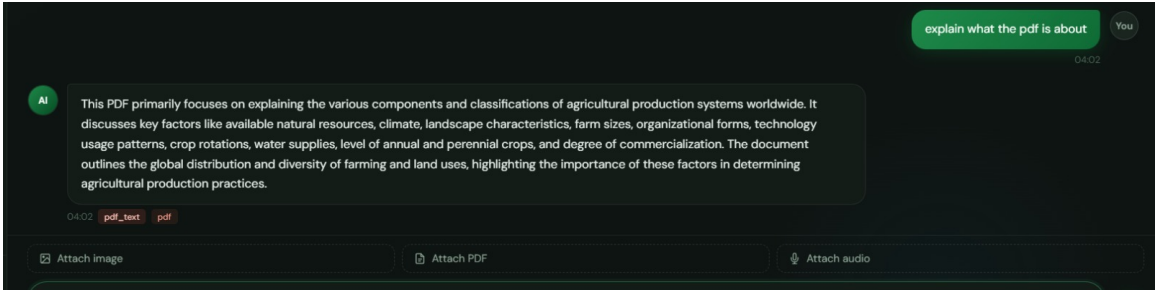


Demo - Image & Text

Demo - Audio


answer
You
04:00
AI
Modern irrigation techniques have significantly improved crop yields and saved water in arctic regions by:
1. Using drip irrigation: This method delivers water directly to the roots of plants at precise times, reducing evaporation.
2. Water-saving sprinklers: These systems use less water than traditional sprinkler systems but still deliver a significant amount of water per application.
3. Precision farming: By using sensors and drones, farmers can monitor soil moisture levels and adjust irrigation accordingly.
4. Drip irrigation
AUDIO TRANSCRIPT
How can modern irrigation techniques improve crop yield while conserving water in arid regions?

Demo - PDF File



Current Limitations

- Multimodal supervision is partly template-generated, so linguistic diversity is limited.
- PlantVillage is curated and clean; field images in real farms may be harder.
- Stage 0 stops when the loss is at about **0.9**, so the language adapter has not yet fully covered the full text dataset.

Next Steps

1. Connect live ASR to the deployed frontend for real spoken queries.
2. Increase Stage 0 training beyond 450 steps and add validation-based early stopping.
3. Validate on real field images beyond PlantVillage.
4. Add multilingual prompts and speech support.
5. Replace template-generated captions and VQA pairs with expert-reviewed supervision.
6. Consider a larger backbone such as `Qwen2.5-1.5B` for stronger reasoning.

Takeaway

Final Message

Agri MLLM is a strong prototype of a **multimodal agricultural assistant**:

- trained first on agricultural text knowledge,
- aligned to **PlantVillage** crop-disease images,
- and presented as a platform that supports **text**, **image + text**, and **audio** interaction.

One assistant, multiple field-friendly ways to ask for help.

References

- Liu, H., Li, C., Li, Y., & Lee, Y. J. (2024). *Improved Baselines with Visual Instruction Tuning*. arXiv:2310.03744. `https://arxiv.org/abs/2310.03744`
- Code repository: `https://github.com/Uness10/HopefullyItWorks`
- Course Lectures